

Railway Transport & Reservation Management System

Phase 3 Submission

Sample Data (INSERT Statements) & Retrieval Queries



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Naming Convention Note

booking.status uses three values: **'CONFIRMED'**, **'WAITING'**, **'CANCELLED'**. This represents the overall state of the booking (PNR level). **ticket_passenger.ticket_status** uses: **'CONFIRMED'**, **'WL'**, **'CANCELLED'**. The shorter **'WL'** label is used at the individual passenger level because it fits the VARCHAR(12) constraint and is consistent with IRCTC-style abbreviations. A booking with status='WAITING' will always have at least one ticket with ticket_status='WL'. This is intentional and clearly documented — not an error.

Part A: Sample Data — INSERT Statements

The database is populated with 12 users (10 passengers + 2 admins), 12 stations, 10 trains, all 6 travel classes, coach/seat hierarchy, 12 bookings spanning confirmed, waiting, and cancelled states, payments via Razorpay/PayU/Paytm, one processed refund, and 8 passenger reviews.

A1 — USER (12 rows)

```
INSERT INTO tts."user" (name, email, password_hash) VALUES
('Khush Hingrajiya', 'khush@gmail.com', 'hashed_khush_001'),
('Dharmesh Upadhyay', 'dharmesh@gmail.com', 'hashed_dhar_002'),
('Vikas Bhabhor', 'vikas@gmail.com', 'hashed_vikas_003'),
('Divya Goti', 'divya@gmail.com', 'hashed_divya_004'),
('Riya Shah', 'riya@gmail.com', 'hashed_riya_005'),
('Amit Patel', 'amit@gmail.com', 'hashed_amit_006'),
('Sneha Joshi', 'sneha@gmail.com', 'hashed_sneha_007'),
('Raj Mehta', 'raj@gmail.com', 'hashed_raj_008'),
('Priya Sharma', 'priya@gmail.com', 'hashed_priya_009'),
('Arjun Verma', 'arjun@gmail.com', 'hashed_arjun_010'),
('Admin One', 'admin1@irctc.in', 'hashed_adm_011'),
('Admin Two', 'admin2@irctc.in', 'hashed_adm_012');
```

A2 — PASSENGER

```
INSERT INTO tts.passenger (user_id, phone, dob, gender) VALUES
(1, '9601824342', '2001-05-14', 'M'), (2, '9824100001', '2001-07-22', 'M'),
(3, '9824100002', '2001-03-11', 'M'), (4, '9824100003', '2001-09-30', 'F'),
(5, '9898000001', '1999-12-01', 'F'), (6, '9898000002', '1998-06-18', 'M'),
(7, '9898000003', '2000-02-25', 'F'), (8, '9898000004', '1997-11-07', 'M'),
(9, '9898000005', '2002-04-19', 'F'), (10, '9898000006', '1995-08-30', 'M');
```

A3 — ADMIN

```
INSERT INTO tts.admin (user_id, employee_id, role) VALUES
(11, 'EMP10011', 'Super Admin'), (12, 'EMP10012', 'Operations Manager');
```

A4 — STATION (12 stations)

```
INSERT INTO tts.station (station_code, station_name, city) VALUES

('ADI','Ahmedabad Junction','Ahmedabad'), ('BRC','Vadodara Junction','Vadodara'),

('ST','Surat Railway Station','Surat'), ('BCT','Mumbai Central','Mumbai'),

('NDLS','New Delhi','New Delhi'), ('JP','Jaipur Junction','Jaipur'),

('AGC','Agra Cantt','Agra'), ('PNBE','Patna Junction','Patna'),

('HWH','Howrah Junction','Kolkata'), ('MAS','Chennai Central','Chennai'),

('SBC','Bengaluru City Junction','Bengaluru'),('PUNE','Pune Junction','Pune');
```

A5 — TRAIN (10 trains)

```
INSERT INTO tts.train (train_number, train_name, source_station_id, dest_station_id) VALUES

(12901,'Gujarat Mail',1,4), (12953,'August Kranti Rajdhani',4,5),

(12931,'Double Decker Express',1,4),(12985,'Rajdhani Express',1,5),

(12009,'Shatabdi Express',1,4), (12952,'Mumbai Rajdhani',5,4),

(12301,'Howrah Rajdhani',5,9), (12621,'Tamil Nadu Express',5,10),

(12028,'Chennai Shatabdi',11,10), (11301,'Udyan Express',11,4);
```

A6 — TRAIN_CLASS (all 6 classes)

```
INSERT INTO tts.train_class (class_code, class_name, base_price) VALUES

('1A','First AC',2500.00), ('2A','Second AC',1500.00), ('3A','Third AC',950.00),

('SL','Sleeper',350.00), ('CC','Chair Car',600.00), ('GEN','General',150.00);
```

A7 — ROUTE_STOP (stops for 8 trains)

```
INSERT INTO tts.route_stop

(train_number,station_id,stop_sequence,arrival_time,departure_time) VALUES

-- Train 12901: ADI → BRC → ST → BCT

(12901,1,1,NULL,'06:00:00'), (12901,2,2,'07:30:00','07:35:00'),

(12901,3,3,'09:15:00','09:20:00'), (12901,4,4,'12:30:00',NULL),

-- Train 12953: BCT → JP → AGC → NDLS

(12953,4,1,NULL,'17:00:00'), (12953,6,2,'03:30:00','03:35:00'),

(12953,7,3,'05:45:00','05:50:00'), (12953,5,4,'08:00:00',NULL),

-- Train 12985: ADI → BRC → BCT → NDLS

(12985,1,1,NULL,'08:00:00'), (12985,2,2,'09:30:00','09:35:00'),

(12985,4,3,'13:00:00','13:15:00'), (12985,5,4,'05:30:00',NULL),

-- Train 12301: NDLS → PNBE → HWH

(12301,5,1,NULL,'16:55:00'), (12301,8,2,'05:00:00','05:10:00'),

(12301,9,3,'09:50:00',NULL);
```

A8 — COACH (26 coaches)

```
INSERT INTO tts.coach (train_number,class_id,coach_number,total_seats) VALUES

-- Train 12901

(12901,1,'H1',24),(12901,2,'A1',46),(12901,2,'A2',46),

(12901,3,'B1',64),(12901,3,'B2',64),

(12901,4,'S1',72),(12901,4,'S2',72),(12901,4,'S3',72),

-- Train 12953

(12953,1,'H1',24),(12953,2,'A1',46),(12953,3,'B1',64),(12953,3,'B2',64),(12953,4,'S1',72),

-- Train 12931 (Chair Car + General)

(12931,5,'C1',78),(12931,5,'C2',78),(12931,6,'G1',90),

-- Train 12985

(12985,1,'H1',24),(12985,2,'A1',46),(12985,3,'B1',64),(12985,4,'S1',72),(12985,4,'S2',72),

-- Train 12301

(12301,1,'H1',24),(12301,2,'A1',46),(12301,3,'B1',64),(12301,4,'S1',72),(12301,4,'S2',72);
```

A9 — SEAT (60 seats across 8 coaches)

```
INSERT INTO tts.seat (coach_id, seat_number, berth_type) VALUES

-- Coach 1 (H1/1A/12901)

(1, '1', 'Lower'), (1, '2', 'Upper'), (1, '3', 'Lower'), (1, '4', 'Upper'),

(1, '5', 'Lower'), (1, '6', 'Upper'), (1, '7', 'Side Lower'), (1, '8', 'Side Upper'),

-- Coach 2 (A1/2A/12901)

(2, '1', 'Lower'), (2, '2', 'Middle'), (2, '3', 'Upper'), (2, '4', 'Lower'),

(2, '5', 'Middle'), (2, '6', 'Upper'), (2, '7', 'Side Lower'), (2, '8', 'Side Upper'),

-- Coach 4 (B1/3A/12901) – seat_ids 17-24

(4, '1', 'Lower'), (4, '2', 'Middle'), (4, '3', 'Upper'), (4, '4', 'Lower'),

(4, '5', 'Middle'), (4, '6', 'Upper'), (4, '7', 'Side Lower'), (4, '8', 'Side Upper'),

-- Coach 6 (S1/SL/12901) – seat_ids 25-32

(6, '1', 'Lower'), (6, '2', 'Middle'), (6, '3', 'Upper'), (6, '4', 'Lower'),

(6, '5', 'Middle'), (6, '6', 'Upper'), (6, '7', 'Side Lower'), (6, '8', 'Side Upper'),

-- Coach 10 (A1/2A/12953) – seat_ids 33-40

(10, '1', 'Lower'), (10, '2', 'Middle'), (10, '3', 'Upper'), (10, '4', 'Lower'),

(10, '5', 'Middle'), (10, '6', 'Upper'), (10, '7', 'Side Lower'), (10, '8', 'Side Upper'),

-- Coach 14 (C1/CC/12931) – seat_ids 41-48

(14, '1', 'Window'), (14, '2', 'Middle'), (14, '3', 'Aisle'), (14, '4', 'Window'),

(14, '5', 'Middle'), (14, '6', 'Aisle'), (14, '7', 'Window'), (14, '8', 'Aisle'),

-- Coach 18 (H1/1A/12985) – seat_ids 49-52

(18, '1', 'Lower'), (18, '2', 'Upper'), (18, '3', 'Lower'), (18, '4', 'Upper'),

-- Coach 19 (A1/2A/12985) – seat_ids 53-56

(19, '1', 'Lower'), (19, '2', 'Upper'), (19, '3', 'Lower'), (19, '4', 'Upper');
```

A10 — BOOKING (12 bookings — confirmed / waiting / cancelled)

```
-- booking.status: 'CONFIRMED' | 'WAITING' | 'CANCELLED'

INSERT INTO tts.booking

(passenger_id,train_number,class_id,source_station_id,dest_station_id,

travel_date,pnr,status,total_fare,booked_at) VALUES

(1,12901,3,1,4,'2026-04-15','PNR1000000001','CONFIRMED', 950.00,'2026-04-01 10:00:00'),

(2,12901,4,1,4,'2026-04-15','PNR1000000002','CONFIRMED', 350.00,'2026-04-01 10:30:00'),

(3,12953,2,4,5,'2026-04-16','PNR1000000003','CONFIRMED',1500.00,'2026-04-02 09:00:00'),

(4,12985,1,1,5,'2026-04-18','PNR1000000004','CONFIRMED',2500.00,'2026-04-02 11:00:00'),

(5,12931,5,1,4,'2026-04-17','PNR1000000005','CONFIRMED', 600.00,'2026-04-03 08:00:00'),

(6,12901,3,1,4,'2026-04-15','PNR1000000006','WAITING', 950.00,'2026-04-03 14:00:00'),

(7,12901,4,2,4,'2026-04-15','PNR1000000007','CONFIRMED', 280.00,'2026-04-04 09:00:00'),

(8,12953,3,4,5,'2026-04-16','PNR1000000008','CANCELLED', 950.00,'2026-04-04 10:00:00'),

(9,12985,2,1,5,'2026-04-18','PNR1000000009','CONFIRMED',3000.00,'2026-04-05 07:30:00'),

(10,12301,4,5,9,'2026-04-20','PNR1000000010','CONFIRMED', 700.00,'2026-04-05 12:00:00'),

(1,12953,2,4,5,'2026-04-22','PNR1000000011','CONFIRMED',1500.00,'2026-04-06 08:00:00'),

(2,12985,3,1,5,'2026-04-25','PNR1000000012','CONFIRMED', 950.00,'2026-04-06 09:00:00');
```

A11 — TICKET_PASSENGER (13 individual travellers)

```
-- ticket_status: 'CONFIRMED' | 'WL' | 'CANCELLED'

INSERT INTO tts.ticket_passenger

(booking_id,seat_id,passenger_name,age,gender,ticket_status,wl_number) VALUES

(1, 17,'Khush Hingrajiya', 22,'M','CONFIRMED',NULL),

(2, 25,'Dharmesh Upadhyay',22,'M','CONFIRMED',NULL),

(3, 33,'Vikas Bhabhor', 22,'M','CONFIRMED',NULL),

(4, 49,'Divya Goti', 22,'F','CONFIRMED',NULL),

(5, 41,'Riya Shah', 24,'F','CONFIRMED',NULL),

(6,NULL,'Amit Patel', 25,'M','WL',1), -- waiting

(7, 26,'Sneha Joshi', 23,'F','CONFIRMED',NULL),

(8,NULL,'Raj Mehta', 26,'M','CANCELLED',NULL),

(9, 53,'Priya Sharma', 24,'F','CONFIRMED',NULL), -- 2-pax booking

(9, 54,'Guest Passenger', 28,'M','CONFIRMED',NULL),

(10,27,'Arjun Verma', 28,'M','CONFIRMED',NULL),

(11,34,'Khush Hingrajiya', 22,'M','CONFIRMED',NULL),

(12,20,'Dharmesh Upadhyay',22,'M','CONFIRMED',NULL);
```

A12 — PAYMENT (12 records via Razorpay / PayU / Paytm)

```
INSERT INTO tts.payment

(booking_id,payment_service,acknowledgment_ref,payment_method,amount,payment_status)

VALUES

(1,'Razorpay','RPY_TXN_A10001','UPI', 950.00,'SUCCESS'),

(2,'Razorpay','RPY_TXN_A10002','UPI', 350.00,'SUCCESS'),

(3,'PayU', 'PAY_TXN_B20001','Net Banking',1500.00,'SUCCESS'),

(4,'Razorpay','RPY_TXN_A10003','Credit Card',2500.00,'SUCCESS'),

(5,'Paytm', 'PTM_TXN_C30001','Wallet', 600.00,'SUCCESS'),

(6,'Razorpay','RPY_TXN_A10004','UPI', 950.00,'PENDING'),

(7,'PayU', 'PAY_TXN_B20002','Debit Card', 280.00,'SUCCESS'),

(8,'Paytm', 'PTM_TXN_C30002','UPI', 950.00,'SUCCESS'),

(9,'Razorpay','RPY_TXN_A10005','Credit Card',3000.00,'SUCCESS'),

(10,'PayU', 'PAY_TXN_B20003','Net Banking', 700.00,'SUCCESS'),

(11,'Razorpay','RPY_TXN_A10006','UPI', 1500.00,'SUCCESS'),

(12,'Paytm', 'PTM_TXN_C30003','Wallet', 950.00,'SUCCESS');
```

A13 — CANCELLATION

```
INSERT INTO tts.cancellation

(booking_id,cancelled_at,refund_amount,refund_status)

VALUES (8,'2026-04-10 15:00:00',712.50,'PROCESSED');
```

A14 — FEEDBACK (8 reviews)

```
INSERT INTO tts.feedback

(passenger_id,booking_id,rating,review,submitted_at) VALUES

(1, 1,5,'Excellent journey, very comfortable!', '2026-04-15 20:00:00'),

(2, 2,4,'Good service, reached on time.', '2026-04-15 21:00:00'),

(3, 3,3,'Average, train was slightly delayed.', '2026-04-16 14:00:00'),

(4, 4,5,'Fantastic! Premium experience all the way.', '2026-04-18 18:00:00'),

(5, 5,4,'Clean coaches and punctual departure.', '2026-04-17 16:00:00'),

(7, 7,4,'Comfortable and affordable.', '2026-04-15 22:00:00'),

(9, 9,5,'Great experience, will travel again.', '2026-04-18 20:00:00'),

(10,10,3,'Could improve food and cleanliness.', '2026-04-20 15:00:00');
```

Part B: Retrieval Queries

17 queries total: 2 simple baseline queries (Q-S1, Q-S2) demonstrating basic SELECT, followed by 15 advanced queries covering every use-case scenario in the project specification. **Fix applied to Q3, Q9, Q14:** the booking LEFT JOIN now always filters by **travel_date AND train_number** to avoid counting tickets from other dates.

#	Type	Use Case	What it answers
Q-S1	Simple	Guest	SELECT * FROM station — list all stations
Q-S2	Simple	Guest	All trains with terminal stations (basic join)
Q1	Complex	Guest	Trains between two stations on a route
Q2	Complex	Guest	Full route schedule with stop timings
Q3	Complex	Guest	Real-time seat availability by class (date-filtered)
Q4	Complex	Passenger	PNR status with seat and WL info
Q5	Complex	Passenger	Complete booking history for a passenger
Q6	Complex	Passenger	Payment details for a booking
Q7	Complex	Admin	Passenger manifest for a train on a date
Q8	Complex	Admin	Class-wise waiting list ordered by WL position
Q9	Complex	Admin	Occupancy % per class per train (date-filtered)
Q10	Complex	Admin	Revenue by train and class (SUCCESS payments only)
Q11	Complex	Admin	Cancellation & refund statistics
Q12	Complex	Admin	Coach types assigned to a train
Q13	Complex	Admin	Cancelled bookings needing WL seat promotion
Q14	Complex	Passenger	Trains with available seats in a class (date-filtered)
Q15	Complex	Both	Average rating and review summary per train

[SIMPLE] Q-S1 — List All Stations

Basic SELECT — shows coverage from simple to advanced queries.

```
-- Q-S1: View all stations in the system

SELECT * FROM tts.station

ORDER BY station_id;
```

[SIMPLE] Q-S2 — List All Trains with Terminal Stations

Basic JOIN — train master data with source/destination station names.

```
-- Q-S2: All trains with terminal station names

SELECT

t.train_number,

t.train_name,

src.station_name AS from_station,

dst.station_name AS to_station

FROM tts.train t

JOIN tts.station src ON src.station_id = t.source_station_id

JOIN tts.station dst ON dst.station_id = t.dest_station_id

ORDER BY t.train_number;
```

Q1 — Trains Between Two Stations

Guest searches for trains from ADI to BCT. Self-join on route_stop ensures the source stop comes before the destination stop in sequence.

```
-- Q1: Trains from Ahmedabad (ADI) to Mumbai (BCT)

SELECT DISTINCT

t.train_number,

t.train_name,

src.station_name AS from_station,

dst.station_name AS to_station,

rs_src.departure_time AS departure,

rs_dst.arrival_time AS arrival

FROM tts.train t

JOIN tts.route_stop rs_src ON rs_src.train_number = t.train_number

JOIN tts.route_stop rs_dst ON rs_dst.train_number = t.train_number

JOIN tts.station src ON src.station_id = rs_src.station_id

JOIN tts.station dst ON dst.station_id = rs_dst.station_id

WHERE src.station_code = 'ADI'

AND dst.station_code = 'BCT'

AND rs_src.stop_sequence < rs_dst.stop_sequence

ORDER BY rs_src.departure_time;
```

Q2 — Full Route Schedule for a Train

Guest views all stops, arrival times, and departure times for train 12901 in order.

```
-- Q2: Complete route of train 12901 (Gujarat Mail)

SELECT

rs.stop_sequence,

s.station_code,

s.station_name,

s.city,

rs.arrival_time,

rs.departure_time

FROM tts.route_stop rs

JOIN tts.station s ON s.station_id = rs.station_id

WHERE rs.train_number = 12901

ORDER BY rs.stop_sequence;
```

Q3 — Real-Time Seat Availability by Class [FIX APPLIED]

Guest checks available seats per class for a specific train on a specific date.

Fix applied: *FIX: The booking LEFT JOIN now filters by BOTH travel_date AND train_number so tickets from other dates are never counted as booked.*

```
-- Q3: Seat availability for train 12901 on 2026-04-15

SELECT
  tc.class_code,
  tc.class_name,
  SUM(c.total_seats) AS total_seats,
  COUNT(tp.ticket_id) FILTER (WHERE tp.ticket_status = 'CONFIRMED')
  AS booked_seats,
  SUM(c.total_seats)
  - COUNT(tp.ticket_id) FILTER (WHERE tp.ticket_status = 'CONFIRMED')
  AS available_seats
FROM tts.coach c
JOIN tts.train_class tc ON tc.class_id = c.class_id
LEFT JOIN tts.seat se ON se.coach_id = c.coach_id
LEFT JOIN tts.ticket_passenger tp ON tp.seat_id = se.seat_id
LEFT JOIN tts.booking b
  ON b.booking_id = tp.booking_id
  AND b.travel_date = '2026-04-15'
  AND b.train_number = 12901
WHERE c.train_number = 12901
GROUP BY tc.class_code, tc.class_name, tc.class_id
ORDER BY tc.class_id;
```

Q4 — PNR Status Lookup

Passenger checks their PNR. Returns all travellers under the booking, their seat assignment (coach-seat or 'Not Assigned' for WL), and ticket status.

```
-- Q4: PNR status for PNR1000000001

SELECT

b.pnr, b.travel_date, t.train_name,
src.station_name AS from_station,
dst.station_name AS to_station,
tc.class_name,
tp.passenger_name, tp.age, tp.gender,
tp.ticket_status, tp.wl_number,
COALESCE(c.coach_number || '-' || se.seat_number,
'Not Assigned') AS seat
FROM tts.booking b
JOIN tts.train t ON t.train_number = b.train_number
JOIN tts.station src ON src.station_id = b.source_station_id
JOIN tts.station dst ON dst.station_id = b.dest_station_id
JOIN tts.train_class tc ON tc.class_id = b.class_id
JOIN tts.ticket_passenger tp ON tp.booking_id = b.booking_id
LEFT JOIN tts.seat se ON se.seat_id = tp.seat_id
LEFT JOIN tts.coach c ON c.coach_id = se.coach_id

WHERE b.pnr = 'PNR1000000001';
```

Q5 — Passenger Booking History

Passenger views all their bookings (past and upcoming) with status and fare details.

```
-- Q5: All bookings for passenger user_id=1 (Khush Hingrajiya)

SELECT

b.booking_id, b.pnr, t.train_name, t.train_number,

src.station_name AS from_station,

dst.station_name AS to_station,

b.travel_date, tc.class_name,

b.status, b.total_fare, b.booked_at

FROM tts.booking b

JOIN tts.train t ON t.train_number = b.train_number

JOIN tts.station src ON src.station_id = b.source_station_id

JOIN tts.station dst ON dst.station_id = b.dest_station_id

JOIN tts.train_class tc ON tc.class_id = b.class_id

WHERE b.passenger_id = 1

ORDER BY b.travel_date DESC;
```

Q6 — Payment Details for a Booking

Passenger retrieves payment service, method, acknowledgment reference, amount and status for a booking.

```
-- Q6: Payment info for PNR 'PNR1000000003'

SELECT

b.pnr, b.total_fare,

p.payment_service, p.payment_method,

p.acknowledgment_ref, p.amount,

p.payment_status, p.paid_at

FROM tts.booking b

JOIN tts.payment p ON p.booking_id = b.booking_id

WHERE b.pnr = 'PNR1000000003';
```

Q7 — Passenger Manifest for a Train and Date

Admin generates the full passenger list for a train on a given date, sorted by coach and seat.

```
-- Q7: Manifest for train 12901 on 2026-04-15

SELECT

b.pnr, u.name AS account_holder,

tp.passenger_name, tp.age, tp.gender,

tc.class_name,

COALESCE(c.coach_number || '-' || se.seat_number,

'WL-' || tp.wl_number::TEXT) AS seat_assignment,

tp.ticket_status

FROM tts.booking b

JOIN tts."user" u ON u.user_id = b.passenger_id

JOIN tts.train_class tc ON tc.class_id = b.class_id

JOIN tts.ticket_passenger tp ON tp.booking_id = b.booking_id

LEFT JOIN tts.seat se ON se.seat_id = tp.seat_id

LEFT JOIN tts.coach c ON c.coach_id = se.coach_id

WHERE b.train_number = 12901 AND b.travel_date = '2026-04-15'

ORDER BY tc.class_id, c.coach_number, se.seat_number;
```

Q8 — Class-Wise Waiting List

Admin sees the WL queue for a specific train and class, ordered by WL position (WL1, WL2, ...).

```
-- Q8: Waiting list for train 12901, class 3A

SELECT

b.pnr, u.name AS passenger_account,

tp.passenger_name, tp.age,

tp.wl_number, b.travel_date, tc.class_name

FROM tts.ticket_passenger tp

JOIN tts.booking b ON b.booking_id = tp.booking_id

JOIN tts."user" u ON u.user_id = b.passenger_id

JOIN tts.train_class tc ON tc.class_id = b.class_id

WHERE b.train_number = 12901

AND b.class_id = 3

AND tp.ticket_status = 'WL'

ORDER BY b.travel_date, tp.wl_number;
```

Q9 — Train Occupancy Report [FIX APPLIED]

Admin views occupancy percentage per class across trains on a given date.

Fix applied: *FIX: booking LEFT JOIN now includes travel_date filter inside the ON clause to prevent cross-date ticket counting.*

```
-- Q9: Occupancy report for all trains on 2026-04-15

SELECT
  t.train_number, t.train_name,
  tc.class_code, tc.class_name,
  SUM(c.total_seats) AS total_seats,
  COUNT(tp.ticket_id) FILTER (WHERE tp.ticket_status = 'CONFIRMED')
  AS confirmed_seats,
  ROUND(
    100.0 * COUNT(tp.ticket_id) FILTER (WHERE tp.ticket_status = 'CONFIRMED')
    / NULLIF(SUM(c.total_seats), 0), 2) AS occupancy_pct
FROM tts.train t
JOIN tts.coach c ON c.train_number = t.train_number
JOIN tts.train_class tc ON tc.class_id = c.class_id
LEFT JOIN tts.seat se ON se.coach_id = c.coach_id
LEFT JOIN tts.ticket_passenger tp ON tp.seat_id = se.seat_id
LEFT JOIN tts.booking b
  ON b.booking_id = tp.booking_id
  AND b.travel_date = '2026-04-15'
GROUP BY t.train_number, t.train_name, tc.class_code, tc.class_name, tc.class_id
ORDER BY t.train_number, tc.class_id;
```

Q10 — Revenue by Train and Class

Admin generates revenue report per train and class. Only counts bookings with a successful payment.

```
-- Q10: Revenue by train and class (SUCCESS payments only)

SELECT

t.train_number, t.train_name,

tc.class_code, tc.class_name,

COUNT(DISTINCT b.booking_id) AS total_bookings,

SUM(p.amount) AS total_revenue

FROM tts.booking b

JOIN tts.train t ON t.train_number = b.train_number

JOIN tts.train_class tc ON tc.class_id = b.class_id

JOIN tts.payment p ON p.booking_id = b.booking_id

WHERE p.payment_status = 'SUCCESS'

GROUP BY t.train_number, t.train_name, tc.class_code, tc.class_name, tc.class_id

ORDER BY t.train_number, tc.class_id;
```

Q11 — Cancellation & Refund Statistics

Admin views cancellation counts and refund processing status per train.

```
-- Q11: Cancellation and refund summary per train

SELECT

t.train_number, t.train_name,

COUNT(can.cancellation_id) AS total_cancellations,

SUM(can.refund_amount) AS total_refunded,

COUNT(*) FILTER (WHERE can.refund_status='PROCESSED') AS refund_processed,

COUNT(*) FILTER (WHERE can.refund_status='PENDING') AS refund_pending,

COUNT(*) FILTER (WHERE can.refund_status='FAILED') AS refund_failed

FROM tts.cancellation can

JOIN tts.booking b ON b.booking_id = can.booking_id

JOIN tts.train t ON t.train_number = b.train_number

GROUP BY t.train_number, t.train_name

ORDER BY total_cancellations DESC;
```


Q12 — Coaches Assigned to a Train

Admin views the full coach configuration for a train — class type, coach number, and seat capacity.

```
-- Q12: Coach configuration for train 12901

SELECT

c.coach_id, c.coach_number,

tc.class_code, tc.class_name,

c.total_seats

FROM tts.coach c

JOIN tts.train_class tc ON tc.class_id = c.class_id

WHERE c.train_number = 12901

ORDER BY tc.class_id, c.coach_number;
```

Q13 — Cancelled Bookings Needing WL Promotion

Admin identifies cancellations where a waiting-list passenger on the same train/class/date should now be confirmed.

```
-- Q13: WL promotion trigger after cancellation

SELECT

b_can.pnr AS cancelled_pnr, b_can.travel_date,

t.train_name, tc.class_name,

b_wl.pnr AS wl_pnr,

tp_wl.passenger_name AS wl_passenger,

tp_wl.wl_number

FROM tts.booking b_can

JOIN tts.train t ON t.train_number = b_can.train_number

JOIN tts.train_class tc ON tc.class_id = b_can.class_id

JOIN tts.cancellation can ON can.booking_id = b_can.booking_id

JOIN tts.booking b_wl

ON b_wl.train_number = b_can.train_number

AND b_wl.class_id = b_can.class_id

AND b_wl.travel_date = b_can.travel_date

AND b_wl.status = 'WAITING'

JOIN tts.ticket_passenger tp_wl

ON tp_wl.booking_id = b_wl.booking_id

AND tp_wl.ticket_status = 'WL'

ORDER BY b_can.travel_date, tp_wl.wl_number;
```

Q14 — Trains with Available Seats in a Specific Class [FIX APPLIED]

Passenger searches for trains from ADI to BCT on a date that still have free 3A seats.

Fix applied: *FIX: booking LEFT JOIN now filters by travel_date inside the ON clause to ensure only same-date tickets are counted.*

```
-- Q14: ADI→BCT on 2026-04-15 with available 3A seats

SELECT
  t.train_number, t.train_name, tc.class_name,
  SUM(c.total_seats) AS total_seats,
  COUNT(tp.ticket_id) FILTER (WHERE tp.ticket_status='CONFIRMED') AS booked,
  SUM(c.total_seats)
  - COUNT(tp.ticket_id) FILTER (WHERE tp.ticket_status='CONFIRMED')
  AS available
FROM tts.train t
JOIN tts.route_stop rs_src ON rs_src.train_number = t.train_number
JOIN tts.route_stop rs_dst ON rs_dst.train_number = t.train_number
JOIN tts.station src ON src.station_id = rs_src.station_id
JOIN tts.station dst ON dst.station_id = rs_dst.station_id
JOIN tts.coach c ON c.train_number = t.train_number
JOIN tts.train_class tc ON tc.class_id = c.class_id
LEFT JOIN tts.seat se ON se.coach_id = c.coach_id
LEFT JOIN tts.ticket_passenger tp ON tp.seat_id = se.seat_id
LEFT JOIN tts.booking b
  ON b.booking_id = tp.booking_id
  AND b.travel_date = '2026-04-15'
WHERE src.station_code = 'ADI' AND dst.station_code = 'BCT'
AND rs_src.stop_sequence < rs_dst.stop_sequence
AND tc.class_code = '3A'
GROUP BY t.train_number, t.train_name, tc.class_name
HAVING SUM(c.total_seats)
  - COUNT(tp.ticket_id) FILTER (WHERE tp.ticket_status='CONFIRMED') > 0
ORDER BY available DESC;
```

Q15 — Feedback & Rating Summary per Train

Passenger or Admin views aggregated ratings and total review count per train, sorted by highest rating.

```
-- Q15: Average rating and review count per train

SELECT

t.train_number, t.train_name,

COUNT(f.feedback_id) AS total_reviews,

ROUND(AVG(f.rating),2) AS avg_rating,

MAX(f.submitted_at) AS latest_review_at

FROM tts.feedback f

JOIN tts.booking b ON b.booking_id = f.booking_id

JOIN tts.train t ON t.train_number = b.train_number

GROUP BY t.train_number, t.train_name

ORDER BY avg_rating DESC;
```

Performance Indexes

Six indexes are created to support the retrieval queries efficiently. The final index **idx_booking_train_date** is new (Suggestion 6) and directly optimises the date-filtered availability queries (Q3, Q9, Q14).

```
-- Existing indexes (from DDL Phase 2)

CREATE INDEX IF NOT EXISTS idx_route_stop_train
ON tts.route_stop(train_number, stop_sequence);

CREATE INDEX IF NOT EXISTS idx_booking_pnr
ON tts.booking(pnr);

CREATE INDEX IF NOT EXISTS idx_booking_passenger
ON tts.booking(passenger_id);

CREATE INDEX IF NOT EXISTS idx_ticket_status
ON tts.ticket_passenger(booking_id, ticket_status);

CREATE INDEX IF NOT EXISTS idx_seat_coach
ON tts.seat(coach_id);

-- NEW: Composite index for availability queries (Suggestion 6)

CREATE INDEX IF NOT EXISTS idx_booking_train_date
ON tts.booking(train_number, travel_date);

-- Directly accelerates Q3 (availability), Q9 (occupancy), Q14 (seat search)
```
